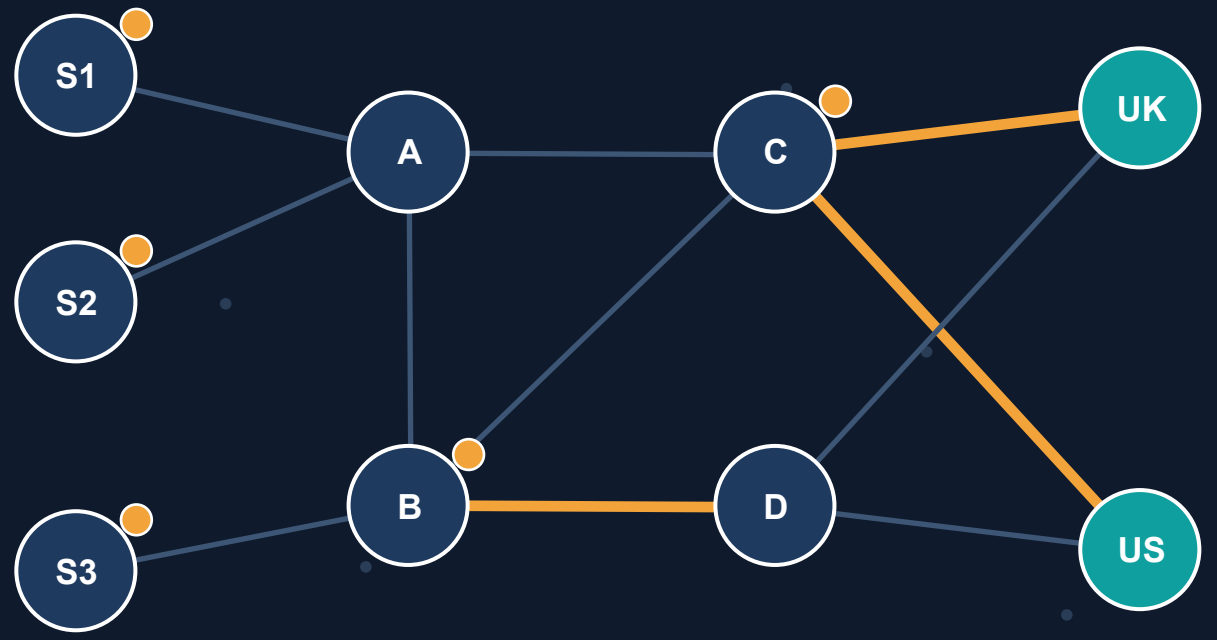
QUVIA · TECHNICAL ASSESSMENT SUBMISSION

Network Throughput Optimization

Latency-aware routing of four traffic streams through a shared, capacity-constrained network

Executive supplement to the analysis notebook



Sarah Reinard Wolberg · **October 2026**

AGENDA

What this supplement covers

01 **Executive summary**
Headline results in one page

02 **Why this matters**
Service, cost and risk at stake

03 **Introduction**
Network, traffic streams, dataset

04 **Problem and objective**
Two-stage lexicographic optimization

05 **Constraints and assumptions**
What the model must respect

06 **Network and naive routing**
Why shortest paths fail

07 **Optimal throughput**
83 of 126 Mbps, and why 83

08 **Split vs. single-path**
Throughput tie, latency difference

09 **Routes by stream**
Paths, delivered and lost, both models

10 **Bottlenecks and stress tests**
Where capacity has value

11 **Fairness**
Who goes unserved at the optimum

12 **Business impact**
From each finding to a decision

13 **Recommendations**
Operating model and investment

14 **Limitations and next steps**
Boundaries and roadmap

15 **Conclusion**
What leadership should take away

The network is capacity-bound at 83 of 126 Mbps

83 Mbps

Maximum throughput

65.9% of 126 Mbps; 43 Mbps cannot be delivered by any routing

3.4×

Naive overload

Independent shortest paths put 61 Mbps on the 18 Mbps B→C link

−24%

Sensitive-latency exposure

Stage 2 cuts 850 → 644 Mbps·ms with no loss of throughput

3 links

True bottlenecks

B→D, C→UK, C→US: +1 Mbps on each lifts throughput to 84

1. **Route jointly.** Per-stream shortest paths are infeasible; a shared-capacity optimizer is required before latency or fairness can be discussed.

2. **Single-path is enough for throughput.** Both models reach 83 Mbps at baseline and in all five stress scenarios. Splitting adds value through lower latency (UK-L 18.3 vs 21.2 ms; US-L 19.6 vs 24.8 ms).

3. **Invest at the cut, not at the busiest link.** Six links hit 100% utilization, but only three add throughput when widened. Together they total exactly 30 + 25 + 28 = 83 Mbps.

4. **Throughput alone is not a service policy.** At the optimum, 7 of 20 source-stream flows receive no service. Fairness and SLA rules should be a defined step in the optimizer.

Every question in the assessment, answered

Assessment question	Answer	Slide
How much of each stream's demand is delivered?	Split: UK-L 16, UK-T 34, US-L 18, US-T 15 Mbps. Single-path: 12, 36, 18, 17 Mbps	12–13
Which path, or split across paths, does each stream use?	Sensitive streams: one fast path per source through C. Tolerant streams use the D corridor and are the only ones that split. 18 / 15 active routes	12–13
How much of each stream cannot be delivered?	Split: 10, 5, 11, 17 Mbps. Single-path: 14, 3, 11, 15 Mbps (UK-L, UK-T, US-L, US-T)	10, 12–13
What is the total delivered across all four streams?	83 of 126 Mbps (65.9%) in both variants, equal to the network's minimum cut	10
Which links are bottlenecks, and what does one more unit add?	B→D, C→UK, C→US: +1 Mbps each (83 → 84). Every other link: 0	14
How should throughput and latency be defined and balanced?	Lexicographic: maximize delivered Mbps, then minimize sensitive Mbps·ms with admissions locked	7
How do the single-path and split variants compare?	Same 83 Mbps. Split lowers sensitive latency: UK-L 18.25 vs 21.17 ms, US-L 19.56 vs 24.78 ms	11
What happens when demand changes?	Re-solved per snapshot: at 60% all 75.6 Mbps delivered; at 150% still 83 Mbps	14

Routing decides service quality, cost and risk

The business question: with finite links, how should traffic be routed, and where should money go, so customers get the most service at the best quality? Each result in this study answers part of that question.

Customer experience

−24%

latency exposure for video and voice traffic, from software alone with no new hardware

Revenue capacity

34%

of today's demand cannot be carried, rising to 56% if demand grows by half

Capital efficiency

3 of 6

saturated links would actually return throughput if upgraded; the other three return 0 Mbps

Service risk

7 of 20

traffic flows get no service at the optimum, and site S3 receives only 8% of its demand

Decisions this analysis supports

1. **How to route today**
   Joint optimizer, single-path by default
2. **Where to invest next**
   B→D, C→UK, C→US, sized in steps
3. **What service policy to set**
   Fairness and SLA floors per class

Four traffic classes compete for one shared network

Two exits (UK, US) receive mixed traffic. Latency-sensitive traffic, such as a video call, loses its value on a slow route. Tolerant traffic, such as a file transfer, only needs to arrive. Both share the same finite links, so routing one stream changes what is left for the others.

Stream	Exit	Latency-sensitive	Demand (Mbps)
UK-L	UK	Yes	26
UK-T	UK	No	39
US-L	US	Yes	29
US-T	US	No	32
Total			**126**

Source-by-stream demand (Mbps), synthetic dataset

Source	UK-L	UK-T	US-L	US-T	Total
S1	6	7	5	6	**24**
S2	5	6	5	5	**21**
S3	5	7	6	6	**24**
B	6	14	8	9	**37**
C	4	5	5	6	**20**

9 nodes

22 directed arcs

20 commodities

68 candidate paths

No production data was supplied, so a synthetic dataset was built with uneven capacities (18–40 Mbps) and latencies (7–18 ms). This creates real fast-but-narrow vs. slow-but-wide tradeoffs. All values are parameters and can be replaced with telemetry without changing the model.

Maximize delivery first, then protect latency

1 Stage 1 · Throughput

Maximize total delivered Mbps, Σ z(s,k), subject to link, exit and demand limits.

Output: admitted amount z*(s,k) per commodity

2 Stage 2 · Latency

Minimize latency exposure Σ $L_p \cdot f_p$ (Mbps·ms) for UK-L and US-L, with every z*(s,k) locked.

Output: the fastest routing of the admitted traffic

Why this design

- **No arbitrary exchange rate.** Lexicographic order avoids inventing a weight between Mbps and milliseconds.
- **Closes a loophole.** Locking each admission stops Stage 2 from "lowering latency" by dropping sensitive traffic and admitting tolerant traffic instead.
- **Two variants solved.** Splittable flow is an LP. Single-path adds binary route choices and becomes a MILP.

Stage 1 core (PuLP, from the notebook)

```
m = LpProblem("Stage1", LpMaximize)
f = {p: LpVariable(f"f_{p}", 0)
     for p in paths}

# objective: total delivered Mbps
m += lpSum(f.values())

# demand: never exceed offer
for k, P_k in paths_of.items():
    m += lpSum(f[p] for p in P_k) \
         <= demand[k]

# capacity: per directed arc
for e, cap in capacity.items():
    m += lpSum(f[p] for p in
         uses[e]) <= cap
```

Every routing plan must respect four hard rules

Constraint	Model form	Business meaning
Link bandwidth	Σ flow on arc e ≤ C_e, per direction	No link is oversubscribed
Exit	UK streams end at UK; US at US	Traffic reaches the right place
Demand	0 ≤ z(s,k) ≤ D(s,k); remainder lost	No phantom traffic; loss is reported
Path rule (split)	Any mix of candidate paths	Maximum routing flexibility
Path rule (single)	Σ y ≤ 1; f ≤ D·y; y binary	One route per source-stream; partial delivery allowed

Single-path linking constraint

```
y = {p: LpVariable(f"y_{p}", cat="Binary") for p in paths}
m += f[p] <= demand[k] * y[p]       # flow only on chosen path
m += lpSum(y[p] for p in P_k) <= 1  # at most one path per (s,k)
```

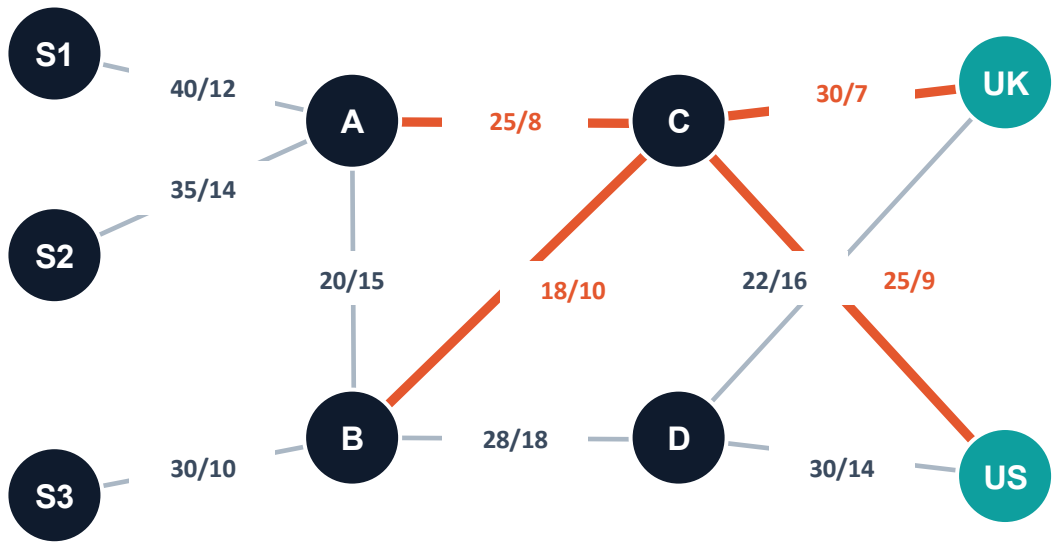
Confirmed with the evaluator

- Demand is known per source node
- Single path applies per source-stream
- Partial delivery is allowed
- Snapshot only: no queuing or carry-over
- Nodes have no capacity limit
- No admission priority for L traffic

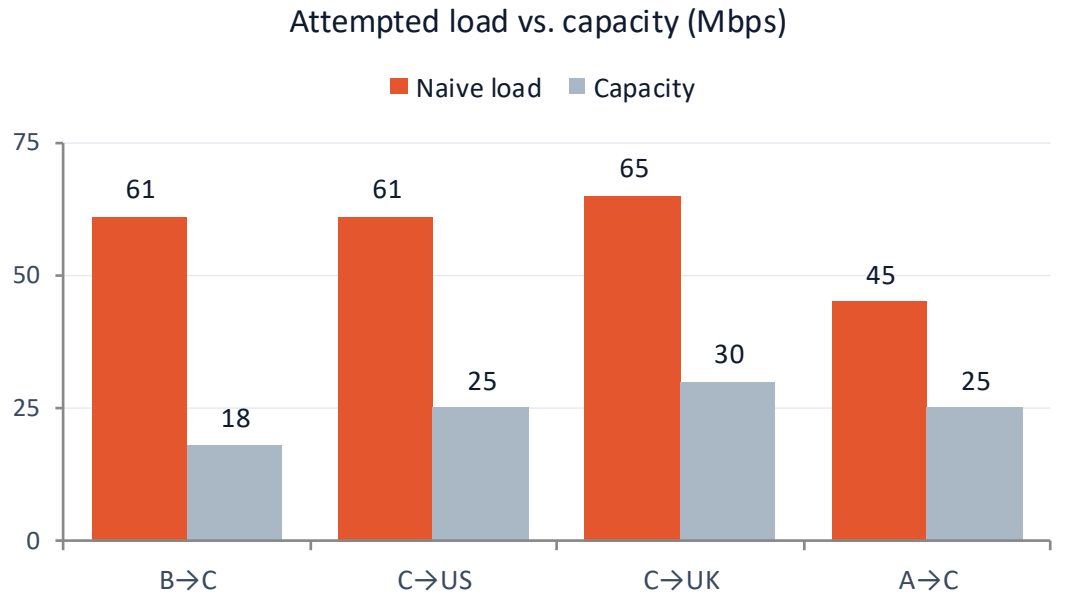
Analyst design choices

- Lexicographic objective
- Flow-weighted latency (worst case reported)
- Synthetic, heterogeneous dataset
- 68 pre-enumerated loop-free paths

Independent shortest paths overload four links



Labels: capacity Mbps / latency ms. **Red** = overloaded under naive routing



Link	Load / Cap	Utilization	Overload
B→C	61 / 18	**339%**	+43
C→US	61 / 25	**244%**	+36
C→UK	65 / 30	**217%**	+35
A→C	45 / 25	**180%**	+20

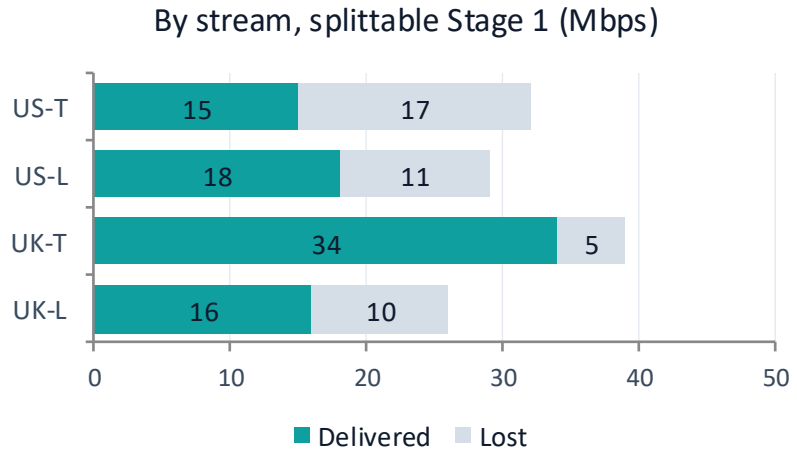
So what: every fastest route runs through C while the D corridor carries 0 Mbps. Without central optimization the network congests and drops traffic it has room to carry.

Optimal routing delivers 83 of 126 Mbps

83 / 126

Mbps delivered at baseline

65.9% delivery · 43 Mbps lost. Same optimum for split and single-path.



Why exactly 83?

Every route to an exit must cross C→UK, C→US or B→D. Their capacities add up to the ceiling:

30 + 25 + 28 = 83

This is the network's minimum cut, so no routing can beat it.

Optimization reroutes traffic onto the idle D corridor

Directed link	Capacity	Naive	Optimized	Change
B→C	18	61	11	-50
C→UK	30	65	30	-35
C→US	25	61	25	-36
B→D	28	0	28	+28
D→UK	22	0	20	+20
D→US	30	0	8	+8

So what for the business

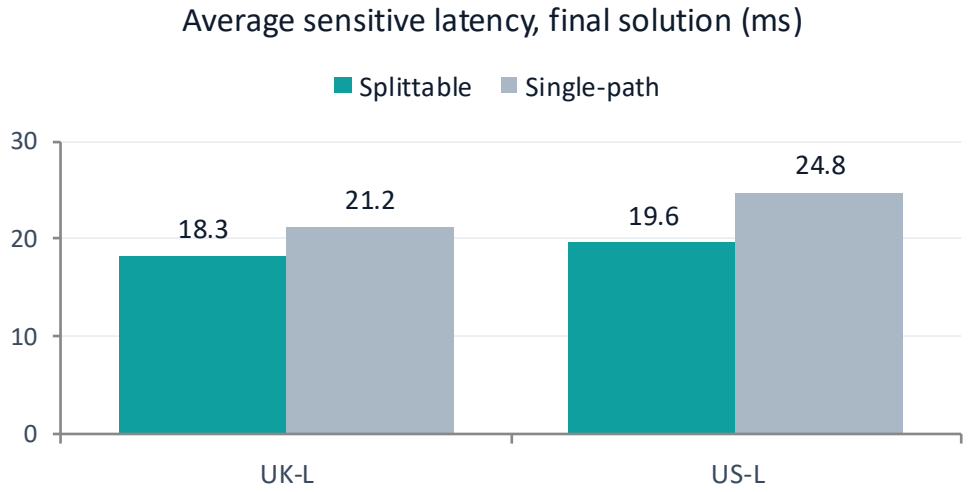
- 83 Mbps becomes the baseline capacity KPI: any policy below it leaves feasible service unused
- Reaching above 83 needs capacity or demand changes, not better routing
- At 150% demand the ceiling is still 83

Single-path ties on throughput; splitting buys latency

Measure	Splittable	Single-path
Maximum throughput	83 Mbps	83 Mbps
Commodities admitted differently	10 of 20	10 of 20
UK-L delivered	16 Mbps	12 Mbps
UK-L avg / worst latency	18.25 / 27 ms	21.17 / 29 ms
US-L avg / worst latency	19.56 / 31 ms	24.78 / 32 ms
Saturated links (final)	6	4
Model size	LP · 68 vars	MILP · 136 vars
Operational complexity	Higher	Lower

So what: operators can choose the simpler single-path design with no capacity penalty, and switch on multipath where faster calls justify the extra complexity.

Caveat: the two pipelines admit different traffic mixes, so the latency gap is a pipeline comparison, not a controlled test of splitting alone.



−24.2%

Stage 2 latency exposure, split: 850 → 644 Mbps·ms at the same 83 Mbps

−14% / −21%

Lower average latency for UK-L / US-L with splitting

Splittable routing: 18 active routes

Stream	Source	Path	Mbps	ms
UK-L	S1	S1 → A → C → UK	**6**	27
UK-L	B	B → C → UK	**6**	17
UK-L	C	C → UK	**4**	7
UK-T	S1	S1 → A → B → C → UK	**4**	44
UK-T	S1	S1 → A → B → D → UK	**3**	61
UK-T	S2	S2 → A → B → D → UK	**5.5**	63
UK-T	S2	S2 → A → C → B → D → UK	**0.5**	66
UK-T	S3	S3 → B → A → C → UK	**2**	40
UK-T	B	B → A → C → UK	**4.5**	30
UK-T	B	B → D → UK	**9.5**	34
UK-T	C	C → UK	**3.5**	7
UK-T	C	C → A → B → D → UK	**1.5**	57
US-L	S2	S2 → A → C → US	**5**	31
US-L	B	B → C → US	**8**	19
US-L	C	C → US	**5**	9
US-T	B	B → A → C → US	**7**	32
US-T	B	B → D → US	**2**	32
US-T	C	C → A → B → D → US	**6**	55

Shaded rows: latency-sensitive streams. ms = path latency.

Per-stream result

Stream	Delivered	Lost	Avg ms
UK-L	16	10	18.25
UK-T	34	5	—
US-L	18	11	19.56
US-T	15	17	—
Total	**83**	**43**	

What the routes show

- **Sensitive traffic stays on one fast path** per source, all through C (7–31 ms)
- **Only tolerant streams split:** 5 commodities use two paths, making room on fast links
- **Some tolerant routes are long** (up to 66 ms), since Stage 2 does not price them; cleanup fixes this

Single-path routing: 15 active routes

Stream	Source	Path	Mbps	ms
UK-L	S2	S2 → A → C → UK	**5**	29
UK-L	B	B → C → UK	**6**	17
UK-L	C	C → UK	**1**	7
UK-T	S1	S1 → A → B → D → UK	**4**	61
UK-T	S2	S2 → A → B → C → UK	**6**	46
UK-T	S3	S3 → B → A → C → UK	**7**	40
UK-T	B	B → D → UK	**14**	34
UK-T	C	C → UK	**5**	7
US-L	S1	S1 → A → C → US	**5**	29
US-L	B	B → D → US	**8**	32
US-L	C	C → US	**5**	9
US-T	S1	S1 → A → B → C → US	**6**	46
US-T	S3	S3 → B → A → C → US	**3**	42
US-T	B	B → D → US	**2**	32
US-T	C	C → US	**6**	9

Shaded rows: latency-sensitive streams. ms = path latency.

Per-stream result

Stream	Delivered	Lost	Avg ms
UK-L	12	14	21.17
UK-T	36	3	—
US-L	18	11	24.78
US-T	17	15	—
Total	**83**	**43**	

What the routes show

- **One route per source-stream:** simpler to configure and monitor
- **Same 83 Mbps, different mix:** UK-L gets 4 Mbps less, UK-T and US-T 2 Mbps more each
- **B's US-L takes slow B → D → US** (32 ms), the main reason US-L latency is higher

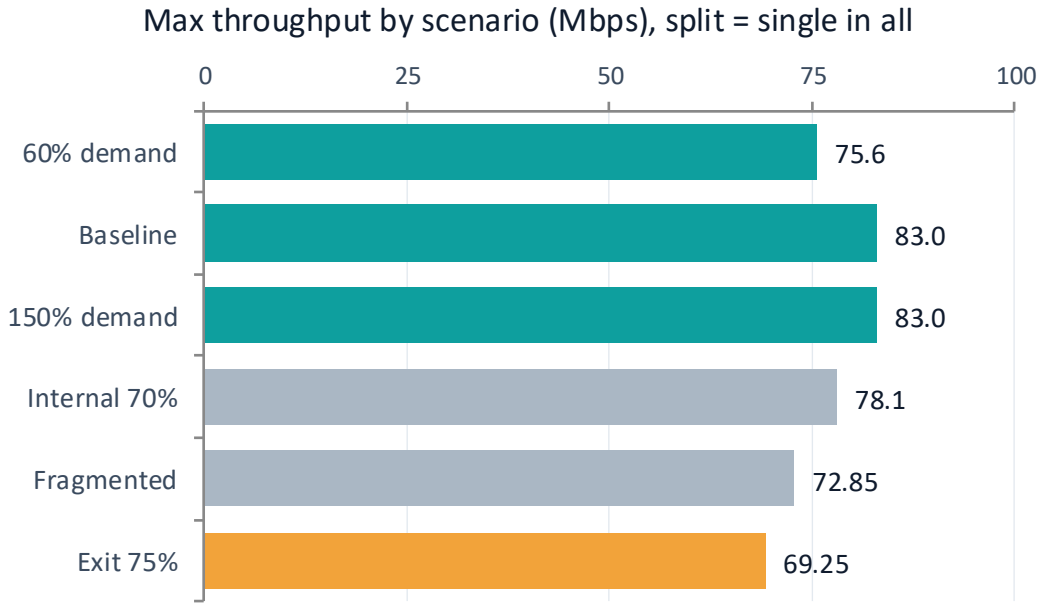
Three links set the ceiling; exit capacity is most fragile

Marginal value test: add +1 Mbps to each of 22 arcs and re-solve

Link	Capacity	Peak util.	Δ throughput	Verdict
B→D	28	100%	**+1.0**	**True bottleneck**
C→UK	30	100%	**+1.0**	**True bottleneck**
C→US	25	100%	**+1.0**	**True bottleneck**
B→C	18	100%	**0**	Busy, no local value
A→C	25	100%	**0**	Busy, no local value
A→B	20	100%	**0**	Busy, no local value
D→UK	22	91%	**0**	Busy, no local value

Saturated ≠ bottleneck. Six links reach 100% in at least one solution, but only the three cut links turn extra bandwidth into delivered traffic (83 → 84 Mbps), and they do so in both models.

So what: an upgrade plan driven by utilization would put half its spend on links that add 0 Mbps. Targeting the three cut links returns 1 Mbps of delivered traffic per Mbps bought.



- **Exit degradation hurts most:** 69.25 Mbps = 22.5 + 18.75 + 28, the same cut at reduced capacity
- **Demand growth is not served:** at 189 Mbps offered, delivery stays at 83 (43.9%)
- **Splitting never adds throughput** in any of the six scenarios

Maximum throughput still leaves some flows unserved

7 of 20

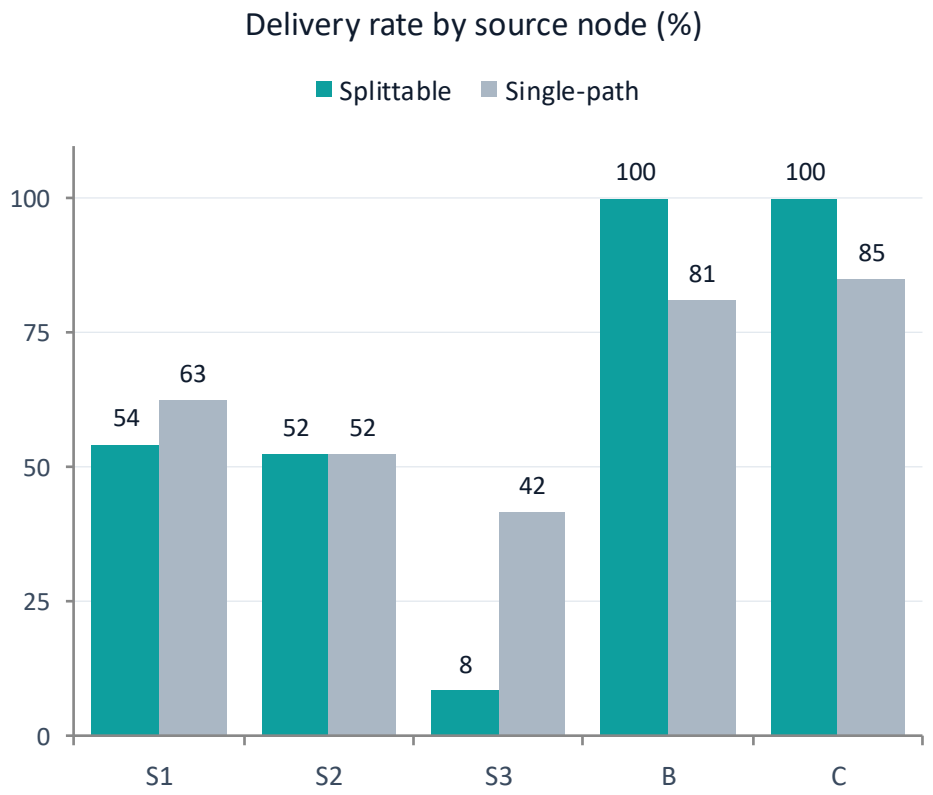
Zero-service flows (split)

Single-path: 5 of 20. Minimum delivery ratio is 0 in both

8%

S3 delivery (split)

S3 gets 2 of 24 Mbps, while B and C, next to the exits, get 100%



Why it happens

Every Mbps counts the same, and many 83 Mbps solutions exist. Traffic that enters near the exits uses the least scarce capacity, so the optimizer serves it first.

Policy fix

Add a step after Stage 1: a minimum delivery ratio, max-min fairness, or SLA floors per class, then minimize latency.

So what: without a stated policy, losses fall on the hardest-to-reach sites, where SLA breaches and churn would appear. One headline KPI can hide this.

The so what: each finding becomes a business decision

Finding	So what	Business impact	Action
Naive routing is infeasible	Local route choices collide on shared links	C corridor at up to 3.4× capacity while the D corridor sits idle: congestion and loss despite spare capacity	Run a joint, capacity-aware optimizer
83 Mbps hard ceiling	Routing cannot create capacity	34% of demand unserved today, 56% at +50% growth; that growth cannot be monetized	Plan capacity and demand policy against the 83 Mbps KPI
Only 3 links have marginal value	Utilization dashboards mislead	3 of 6 saturated links return 0 Mbps if upgraded; the cut links return 1:1	Direct capex to B→D, C→UK, C→US; re-optimize after each step
Stage 2 and splitting cut latency	Quality improves without new capacity	−24% sensitive latency exposure; UK-L/US-L averages −14%/−21% at the same 83 Mbps	Keep multipath for latency-critical classes
Single-path matches throughput	Simplicity costs nothing at baseline	15 vs. 18 routes, deterministic paths, easier troubleshooting	Default to single-path routing
7 of 20 flows unserved	Max throughput is not fair service	Losses concentrate on specific sites (S3 at 8%): SLA and churn risk	Add a fairness / SLA step
Exit degradation hurts most	Exits are the fragile point	25% exit loss costs 13.75 Mbps, nearly 3× the internal-corridor impact	Prioritize exit redundancy and monitoring

Route jointly, set policy, then invest at the cut

Recommended optimization stack

1. **Throughput**
   Maximize delivered Mbps (83 at baseline)
2. **Fairness / SLA**
   Apply minimum service floors and class rules
3. **Sensitive latency**
   Minimize UK-L / US-L exposure, admissions locked
4. **Cleanup**
   Shorten tolerant routes; reduce active paths

Step 2 is new: it closes the starvation gap shown on slide 15.

Operations

Default to single-path routing: same throughput, simpler to run and troubleshoot. Turn on multipath for latency-critical classes or under stress.

Investment

Prioritize B→D, C→UK and C→US. Size upgrades by re-optimizing in steps, since the next cut takes over once one binds.

Measurement

Swap utilization dashboards for marginal-value analysis. Track delivery and latency by source and class, not just total Mbps.

Demand

With the ceiling fixed at 83 Mbps, growth needs capacity, admission policy or demand shaping. Routing alone cannot absorb it.

Results are rigorous inside clear boundaries

Limitation	Effect on the results
Synthetic data	83 Mbps is specific to this network; the method transfers
Fixed link latency	No queuing delay, so latency near 100% utilization is optimistic
Local marginal value	+1 Mbps value does not scale linearly; joint upgrades not tested
Pre-enumerated paths	Optimal over 68 candidates; larger networks need path generation
Different admission mixes	Split vs. single latency gap is not fully controlled
Deterministic snapshot	No uncertainty, failure probabilities or route-churn cost

Roadmap

P1 — **Fairness and SLA constraints**
Minimum delivery ratios, class floors

P2 — **Telemetry calibration**
Real capacity, latency and demand

P3 — **Queue-aware latency**
Delay that rises with utilization

P4 — **Throughput–latency frontier**
Show leaders the tradeoff curve

P5 — **Scale and resilience**
Column generation, failure scenarios

CONCLUSION

The network is capacity-bound, not routing-bound

- **83 Mbps** is a hard ceiling set by three links. Optimization reaches it; only capacity or policy moves it.
- **Joint routing** is mandatory. Independent shortest paths overload links by up to 3.4×.
- **Single-path** is sufficient for throughput. Splitting is worth keeping for latency-critical traffic.
- **Fairness** must be designed in. The throughput optimum can leave entire flows unserved.

So what: better routing is a free, immediate gain; serving more customers needs targeted capacity at three links; protecting every customer needs an explicit fairness policy.

Thank you · Questions welcome

Full analysis: Network_Routing_Problem.ipynb and PDF

Two-stage model in PuLP (CBC solver)

Stage 2: minimize sensitive-latency exposure with admissions locked

```
m2 = LpProblem("Stage2", LpMinimize)

# Mbps x ms, sensitive streams only
m2 += lpSum(latency[p] * f[p]
            for p in paths_L)

# lock each commodity's Stage 1 result
for k, P_k in paths_of.items():
    m2 += lpSum(f[p] for p in P_k) == z_star[k]

# + same capacity (and single-path) rules
```

Validation built into the notebook

- Demand reconciles: 126 Mbps across 20 commodities
- No arc exceeds capacity; all solves report Optimal
- Single ≤ split holds in every scenario; Stage 2 keeps 83 Mbps

Notation	Meaning
D(s,k)	Offered demand, source s, stream k
z(s,k)	Delivered amount (Stage 1 decision)
f_p	Mbps carried on candidate path p
y_p	1 if path p is selected (single-path)
C_e, L_e	Capacity and latency of directed arc e
$L_p = \Sigma L_e$	Path latency, sum over its arcs
MV_e	$T^*(C_e + 1) - T^*(C_e)$, marginal value

Problem sizes: split LP 68 continuous variables; single-path MILP adds 68 binaries (136 total). The model is layout-agnostic: the 8-node hub variant runs by swapping the link table.